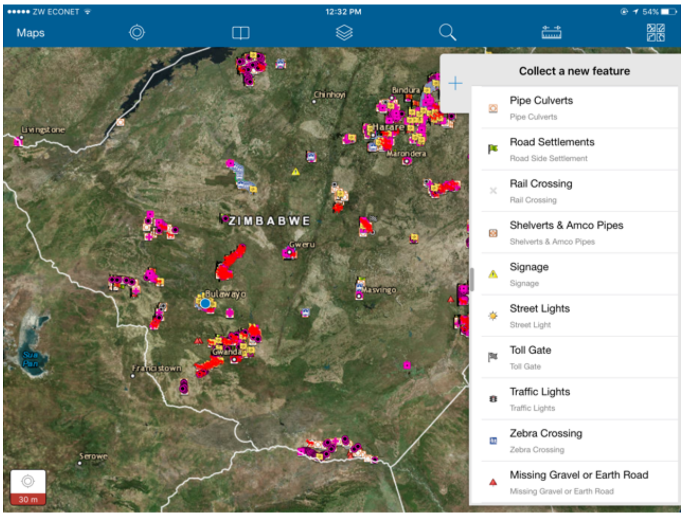
ROAD CONDITION & INVENTORY SURVEYS

ASSESSMENT MANUAL
(Draft rev 1)

For the
DEPARTMENT OF ROADS,RURAL DISTRICT COUNCILS , URBAN COUNCILS & DISTRICT DEVELOPMENT FUND



Document Reference

RCS/01/MN/ 01

Date Of Issue

10 October 2016

Compiled by Zimbabwe Local Government Association (ZILGA)

Contents

Glossary of Acronyms

ARDCZ – Association of Rural District Councils of Zimbabwe
CBD – Central Business District
DDF – District Development Fund
DoR – Department of Roads
GIS – Geographical Information Systems
NRCS – National Road Condition Survey
PPE – Personal Protective Equipment
RDC(s) – Rural District Council(s)
RMU- Road Maintenance Unit
RDB – Roads Database
RRA – Rural Road Audit
UCs - Urban Councils
VCI- Visual Condition Index
ZINARA- Zimbabwe National Roads Administration
ZILGA – Zimbabwe Local Government Association

1 INTRODUCTION

1.1 Purpose & Background

This document provides guidelines for carrying out visual road condition and inventory surveys.

Visual evaluations can be used for determining:

- Visual Condition indices
- Maintenance & Rehabilitation Needs
- Establishment of a Roads Network Database; and
- Prioritisation at network level

The Manual is for Visual Assessors and should be used for the training of the Assessors.

1.2 Layout of the Manual

The Manual is Layout Out is as Follows:

- Part 1 - This Manual
- Part II – Annexures to Part 1 containing photos plates and rating examples
- Part III – Data Recording using ArcGIS Collector

2 GENERAL INFORMATION

2.1 The Roads Acts (13:38)

The Roads Act [Chapter 13:18] was enacted in 2001. It sets out the legal and institutional framework for the planning, development, construction and maintenance of the Zimbabwe's public road network. It supersedes the previous Road Act [Chapter 13:12] from the pre-Independence era. The purpose of the Act was to bring into being establish the Zimbabwe National Road Administration (ZINARA) and the Road Fund.

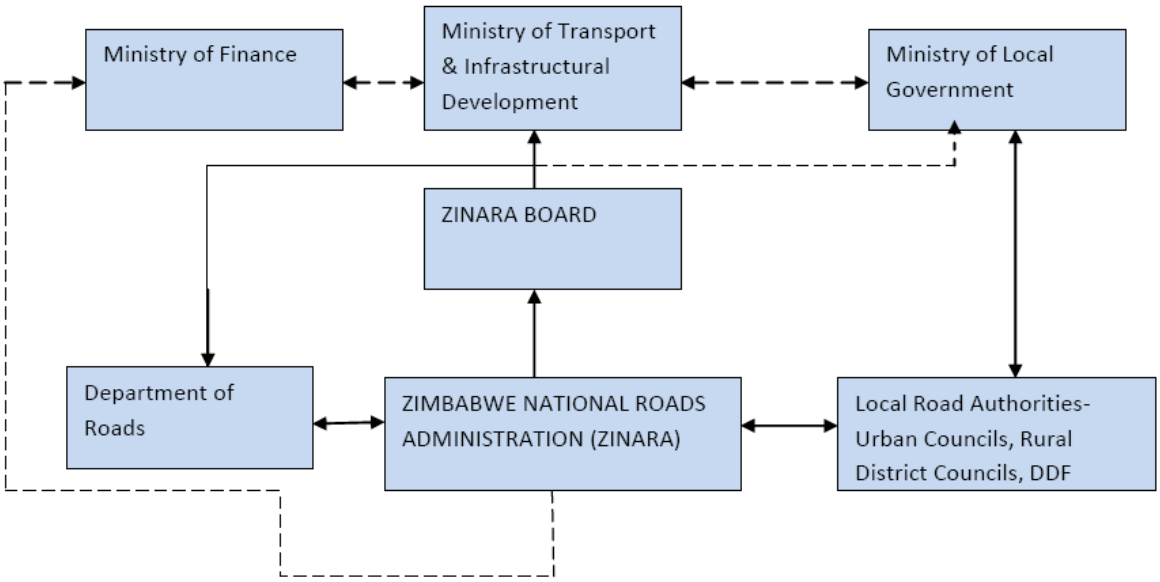
The Act also provides for road authorities and their functions, and for the planning, development, construction and maintenance of the road network.
This includes the regulation of standards, classification of roads, safety and environmental considerations, control of entry upon roads, and the acquisition of land and materials for road works. The Roads Act covers all regional, primary, secondary, tertiary and urban roads in Zimbabwe, except those in the national parks and wildlife estates. The Road Act was amended through the General Laws Amendment No. 2 o f 2002. This amendment included two important new measures affecting the road sector: 1) the District Development Fund was nominated as a Road Authority, and 2) the definition of "road user charges" (due to the Road Fund) was changed to include all fees charged and collected in terms o f the Vehicle Registration and Licensing Act. (Previously the Road Fund could only collect vehicle license fees).

2.2 Overall Network Size

The Country's estimated total road network of 88,330km is managed by the following road Authorities:-

- The Department of Roads in the Ministry of Transport Infrastructural Development (18,430km);
- Urban Councils (UC) (8,530km) for Urban roads;
- The District Development Fund (DDF) in liaison with local authorities (26,000km);
- Rural District Councils (RDC) (35,370km)

2.3 Institutional Set-up Structure



2.4 Functions of Road Authorities (Roads Act(13:18)

- Planning, design, construction, maintenance, rehabilitation and management of roads
- Set operational priorities
- Award contracts for design construction, maintenance, rehabilitation and management of roads
- **Establish and maintain an information management system concerning roads**

3 DEFINITIONS

3.1 Types of roads

Sealed Roads – a paved road

Subtypes of sealed roads

Narrow Mat **–** a paved road where the width of the paved section is less than the standard 7m

Wide Mat – width of the paved section is 7m i.e. 2 lanes each of 3.5m wide.

Wide Mat Gravel Shoulders – similar to side mate but with gravel shoulders

Wide Mat Surfaced Shoulders – similar to wide mat but with surfaces shoulders.

Strip – the paved section of the road are on the wheel tracks.

Gravel Roads - an unpaved road with gravel as the wearing coarse

Earth Roads –an unpaved road with no gravel or wearing course

Another term commonly used by the common man in the street is “dust “roads. Again, these are both gravel and earth roads.

3.2 Classes of Roads

Zimbabwe’s Roads are classified according to the service (mobility, access or both) requirement of the road, herein called Functional Classification[1].

Regional road” means any road forming part of the Regional Trunk Road Network which has been declared to be a regional road in terms of this Act;

Regional Trunk Road Network” means roads linking countries within the Southern African region;

Primary road” means a road not forming part of the Regional Trunk Road Network which links regional roads to urban centres or urban centres to each other, and which has been declared to be a primary road under Roads Act;

[1] Roads Re-classification Manual,1999

Secondary road" means a road which connects regional, primary, tertiary and urban roads, industrial and mining centres, tourist attractions and minor border posts to each other and which has been declared to be a secondary road in terms of the Roads Act;

Tertiary road" means a road which provides access to schools, health centres, dip tanks and other service facilities within a rural district council area or connects and provides access to secondary, primary and regional roads within and outside a rural district council area and which has been declared to be a tertiary road in terms of the Roads Act;

Sub-classes of Tertiary Roads

Tertiary Feeder - Roads that connect higher order roads to rural service centres in the district or to smaller isolated mines or
connect secondary points (e.g. minor border posts, growth points, industrial areas, tourist attractions to rural service centres in the district or connect rural service centres to each other.

Tertiary Access – Access roads that serve schools, clinics, health centres dip tanks and other service facilities, residential areas and farms.

Urban road" means

(a) any road within an urban council area, other than a secondary, primary or regional road;
(b) any road located on urban land in a rural district council, other than a tertiary, secondary, primary or regional road; which has been declared to be an urban road in terms of the Roads Act.

Sub-Classes of Urban Roads

Arterial – primary internal network of an urban or surbuban area. Their function is for long distance travel within the urban area. They generally carry high traffic volumes.

Distributor – Long distance arterials which distribute traffic over wide areas. Although all roads have a "distribution function"

Collector – Streets that carry the main internal traffic movement within a limited portion of time and serve as feeder to the arterial roads.

Local – streets are primarily for access to abutting properties and only carry low volume traffic

CBD Roads – Generally carry high traffic volumes which include a significant proportion of short distance trips. The location of CBD roads are designated by the urban authority.

Industrial Roads - These are roads within the Industrial Areas

3.3 Ownership of roads by Class

Road Authority	Road Class
Department of Roads	Regional Roads, Primary Roads Secondary roads
Rural District Council (RDCs)	Tertiary roads, Urban roads
Urban Councils (Municipality, Town Council or Local Board)	Urban roads
District Development Fund(DDF)	Urban Roads & Tertiary Roads which were constructed, maintained or rehabilitated by the District Development Fund within the area of jurisdiction of any local authority immediately before the date of commencement of this Act, the District Development Fund;
Private Individuals or Entities	Other roads

3.4 Road Segmentation/Assessment Segment

A Road Segment or Assessment Segment is the length of road for which one assessment rating is recorded. The following parameters will be used to define a segment:-

- Length of road;

- Traffic
- Boundaries (province, district, administrative)

Type of Road	Recommended Segment Length(Km)		
	Minimum	Standard	Maximum
Sealed &Gravel Roads in Rural Areas	2.5	5	7.5
Earth Roads	5	10	15
Roads in Peri-Urban and Freeways	1	2	3
Urban Roads	0.25	0.5	0.75

Table xx: Recommended Segment Length

Example of Calculating Segment Lengths

1. Assume a rural road being 17 km long. The Segment lengths will be as follows: Segment 1 = 5km; Segment 2 = 5km and Segment 3 = 7km (and not 5km and then 2km – since 2km is less than the minimum 2.5km)
2. Assume a rural road being 14km long. The Segment lengths will be as follows: Segment 1 =5km; Segment 2 = 5km; and Segment 3 = 4km (since this is greater than 2.5km)
3. Roads whose total length is less than the minimum segment length should be assessed as 1 Segment. For example a route which is ,say 1.2km will be taken as one Segment.

Roads Database
All segments in the network should be identifiable on the Road DataBase.
The direction in which kilometres should be recorded should also confirm with the RDB. Whenever possible the same segmentation should be used in the next survey.

3.5 Road Inventory

Road Inventories are one of the important support tools in the management of road network maintenance and other planning. They can be in written or computerised form. Features of a well organized road inventories are:-

- Full specification of the locations of the roads
- Data associated with a road length or chainage
- Data grouped for ease of use according to purpose
- Regular surveys to update the data where appropriate
- Updated regularly in order to include new roads
- A variety of report formats available (computerised inventories)

3.6 Evaluation of Condition of Road Pavement

There are generally two points of view when assessing the conditions of roads, namely(1) the Road User point of view and (2) the Road Engineer Point of view as described below:-

1. The **road user** views the road as a “service”. He values the road in terms of those parameters which affect quality of travel.

- Comfort
- Safety
- Vehicle operating costs

2. The **road engineer** who values the road as a ‘load bearing structure’ to be maintained in good time for it to remain serviceable at optimum costs; he also recognises the functional requirements.

According to the Department of Roads Visual Condition Survey Manual (1995), A *Road Condition Survey* is an exercise which is carried out to ascertain the condition of roads in the network.

An alternative to this basic method is to employ relatively advanced and complex technical equipment capable of automatically measuring certain characteristics. The high cost of this type of inspection, however, makes it not viable for most of the roads in many districts in Zimbabwe.

The condition survey is generally by means of visual distress assessment by trained Assessors. The distress is described by recording its main characteristics, namely:-

- **Type** – for example roughness, rutting, edge breaks, gravel thickness, potholes
- **Degree of severity** – how bad is the condition?
- **Extent of occurrence** – how widespread is the condition,

With this information, a Visual Condition Index (VCI) can be calculated to give an overall rating of the condition of the road segment.

Degree of severity

The degree of a particular distress is a measure of its severity. Since the degree on a given segment can vary over the segment length, the degree to be recorded should, in connection with the extent of occurrence, give **the best average assessment of the seriousness** of the distress over the segment.

Extent

The extent of distress gives a measure of how widespread the distress occurs over the segment length.

Extent	Description
0-25%	Isolated occurrence, not representative of the segment length being evaluated (seldom)
25-50%	Intermittent (scattered) occurrence, over parts of the segment length (more than isolated)
51-75%	Intermittent(scattered) occurrence, over most parts of the segment length(general, or extensive occurrence over a limited portion of the segment length.
75% -100%	More frequent occurrence over a major portion of the segment length, and extensive occurrence.

3.7 Visual Condition Index

The condition index can be used;

1. To give an indication of pavement condition for each segment
2. To indicate the changes in the condition of a pavement over time and;
3. To classify the road into one of the following condition categories.

VCI CATEGORY	**POOR**	**FAIR**	**GOOD**	**VERY GOOD**
	0 - 30	**30 - 55**	**55 - 80**	**80 - 100**
Where VCI stands for 'Visual Condition Index'				

Table 1: Visual Condition Index[2]

The VCI can then be used to identify **maintenance** and rehabilitation measures and priorities. These identified needs are generally not for use at Project Level (Implementation), but are generally used as input for programming and budgeting at network level.

[2] Visual Condition Survey Manual, Department of Roads, Zimbabwe,1995

4 ASSESSMENT PROCEDURE & QUALITY ASSURANCE

4.1 Team Structure

The team will be composed of a Team Coordinator /Team Leader, two Assessors and a Driver. The roles and responsibilities of the each member of the team is given below:-

The Team Coordinator – is expected to be an Engineer or Professional within the District /Road Authority with some degree of responsibility to secure and allocate resources and superintend the use of those resources. He must also be able produce reports on programme and progress. It is important that the Team Coordinator take part in the field survey to ensure that he/she is familiar with the methodology and have a feel of the challenges the team is working with.

Three (3) people are expected to be in the field at any time (Driver and 2 Assessors or Driver, Assessor and Engineer). This is necessary in order to improve the safety of the team on the road. The team must among themselves know the road network to a considerable extent.

Driver – must know the road network, act as a Safety Officer sentinel to ensure the safety of the Assessors doing the measuring and data input.

The two Assessors concentrate on observing distresses on the road, road features and input their observations into a pre-fined form on the IPADS.

4.2 Training of Visual Assessors

The accuracy of the visual assessments depends largely on the knowledge, experience and commitment of the visual assessors. To minimize the element of subjectivity and to ensure good knowledge of the assessment procedure, it is absolutely essential that assessors are trained prior to carrying out the field work. Three (3)Assessors per District (1 DDF, 1 DoR and 1 RDC) are being trained to ensure provide redundancy, provide relief and ensure that in the event that the survey can continue in the event of absence of one of the Assessor. It is recommended that an Assessor take a one week break after 2 weeks of continuous work in the field to minimize fatigue.

4.3 Procedure for Visual Assessment

The first stop should be made at the first km peg of the segment and the second at the last km peg, this will allow the assessor to adjust to the prevailing conditions and thus calibrate his vision. Additional stops should be made where change occurs, and when the team encounters road features like culverts, bridges and sign . Every visible distress on the segment must be taken into account in summary.

At the end of each segment, assessors should record their overall ratings before moving to the next segment.
It is advisable that the Assessors have a note book in which they can jot their observations at say every km to assist them in coming up with a summary at the end of the Segment.

For a good quality assessment, the length of road should not exceed **80km per day for rural roads and 60km for urban and peri urban roads**. The best results are obtained when the assessors are driving at a speed of less than 20km/hr on the shoulder if on major highways.

4.4 Field Checking

A representative sample (say approximately 10%) of all roads assessed will be checked independently to confirm correctness of the assessments made.

4.5 Planning Logistics

- Tools & Equipment for the survey – vehicles, measuring tapes, personal Protective Equipment (PPE) etc
- Route Mapping -_necessary to ensure that mileage is reduced. The Team must identify /plan in advance where the team(s) will put-up for the night.
- Duration timing – Each team is expected to do a minimum of 80km per day rural roads, 60km per day highways, 40km per day –urban roads
- Data will be stored on Ipads and transmitted as soon as the team returns to base.

4.6 Vehicles, Tools & Equipment Requirements

Item No.	Item Description	Qty – per team
1	**VEHICLES** 4X4 Vehicle in good condition, twin cab or similar Fitted with Safety Signs and revolving Amber Light.	1
2	Fuel requirements – (Assumed 10km per litre)	400ltrs
	Team 1: T & S Allowance Technician, Resource persons Driver for 3 people for 20 days	80 mandays
6	**MEASURING TAPES**	
	Measuring wheel	1
	5m tape	1
	ruler 30cm	1
	Safety vests	3
	Revolving Beacon light	1

4.7 Safety & Health Considerations

4.7.1 Overview

The survey is happening in a live traffic environment. It is critical that the Survey Team exercises due care to ensure the Safety of the team and the motoring public. Listed below are some of the common hazards and possible mitigation measures to reduce the risks.

4.7.2 Notifications to Local Police

It is critical that the Local Police must also be notified prior to going out in the field so that they are aware that such an exercise is happening in their area.

4.7.3 Risk & Mitigation Measures

Item No.	Description of Hazard Condition or Risk	Mitigation Measures
	Vehicle breakdown	1. Carry out Tool box talks 2. Have Mobile phone with airtime 3. Use reliable vehicle 4. Part of team to have mechanic experience
	Wild animal attack/including snakes/bees	5. Check surrounding before disembark 6. Cautions (always on look-out) – especially 7. Wear protective clothing: safety shoes, long pants 8. Prior knowledge of wildlife areas 9. Do work in wildlife areas during daylight
	Malfunction of Hardware/software (due to physical damage like dust)	10. Always have protective cover on IPAD on. 11. Due care during stopping and taking off 12. Use it for exercise only – no downloading unnecessary information. 13. Disable automatic updates function
	Team member gets sick/malaria	14. More than two team members trained 15. Mosquito repellent: first aid kit
	Vehicle Accident: other drivers (reckless), donkeys, cows, speed.	16. Exercise caution 17. Do not drink and drive 18. Speed 19. Wear a reflective vest 20. Use flashing lights on the car 21. Clear sight times at stoppage points 22. Stand, stop, pull off road 23. Safety briefings by team leaders
	Fall from height (off bridges)	Do not lean on bridge parapets or guardrails.
	Landmines Are a common occurrence in some districts. Please identify the routes where this is so and come up with a risk assessment prior to doing the survey.	

5 FEATURES & DISTRESS CONDITION SURVEY DATA COLLECTION

5.1 General

The customized and standardized data collection for the Road Condition & Inventory Survey is detailed under the following 12 feature classes as given in Table xx below:

Generally the data is categorized as follows:-

5.2 Feature Classes

Item #	Description	Category		
		Mandatory	Essential	Desirable
1	Sealed Roads			
2	Gravel Roads			
3	Earth Roads			
4	Bridges			
5	Pipe Culverts			
6	Box Culverts			
7	Shelverts & Armco Pipes			
8	Piped Causeways			
9	Drift			
10	Road Signs (Signage)			
11	Junctions			
12	Cattle Grids (Grid)			
13	Impassible Section			
14	Traffic Lights			
15	Street Lights			
16	Bus Stops			
17	Laybys			
18	Rail Crossing			
19	Tollgates			
20	Footbridges			
21	Missing Sealed Road			
22	Missing Gravel Road			
23	Settlements			
24	Catchpit			
25	General Point			

Basic Data – containing information describing/ identifying road i.e. Road Name, Class, Route Number, climatic region, Authority Name and others;

Geometric Data – contain information on the dimensions of the road or road feature i.e. length ,width height (where applicable)

Condition Serviceability Data – containing information on the condition of the road or the road feature.

Traffic Data – Information relating to the volume and type of traffic if available needs to be provided.

Design & Construction Data – containing information of when the road or road feature was constructed, dates of the previous re-gravelling or resealing etc.

5.3 Road Segment Information - Basic Data

Item #	Description	Category		
		Mandatory	Essential	Desirable
1	Route Number	yes		
2	Road Name		yes	
3	Road Class	yes		
8	Road Type	yes		
4	Chainage from(km)		yes	
5	Chainage to (km)		yes	
6	Segment Length (m)		yes	
7	Segment width (m)	yes		
9	Datum Point	yes		
10	Authority Name	yes		
11	Climatic Region		yes	
12	Terrain Type		yes	

Route Number (Essential)

Route Number must be entered on ALL road segments and road features (e.g. Bridges, Signs, Culverts etc).

Numbering format – District code as per the Registrar General Code – 50S1/1 (Mutasa Sealed Road 1 Segment 1); 50G1/12 ; 50E1/2

Existing route number – use this if available, otherwise use the numbering format suggested above. for new roads

If a **dual carriageway** is evaluated, each carriageway should be evaluated separately. The two carriageway segments will have the same road number and kilometer distances. The direction of travel relative to the distance markers should be used to indicate which of the two sections is evaluated. For example, “N” after the road number would indicate that the traffic on this particular carriageway is travelling northwards.”E”, “S”, and “W” would similarly indicate East, South and West respectively

The item should cover the number of the road and section concerned , for example:-

50S1/1E means route number 50S1, section 1 and Eastbound on dual carriageway.

43S1/12 N Means route number N43S1, section/segment 12 and Northbound carriageway.

Road Name (Desirable)

This is the road name as decided or used by the Road Authority

Road Class (Essential)

Classify the road using the definitions provided in section 3.2 of this Manual. This can be done in the field or during a review of the day’s work. In situations where the Road Segments are more than 3, you must enter Road Class on the first and Last Segments , and may skip entering this field on intermediate segments.

Road Type (Mandatory)

Indicate the road type using the definitions provided in section 3.1 of this manual. Enter this information for ALL Road Segments.

Datum Point (Mandatory)

The datum or start point must be defined/described –e.g. if a road starts from a another road say route 50S1 (Mashongwe Rd) and end on Route 50G15 Thurlows Road, then the datum point may either be 50S1 or 50G1 . This is necessary so that the same datum point can be used in future repeat exercises.

Only on first and last segments.

Authority Name (mandatory)

This is the Road Authority responsible for the road and or road segment - for example, if the road segment is owned by DDF, then the Authority Name is DDF.

Terrain Type (Mandatory)

The terrain type is used for calculating excess user costs, and one of the types must be selected from the drop down list. The Terrain type is defined by the gradient & or curvature, the worst case must always be selected.

Terrain Type	Gradient	Curvature
Flat	Gradient mostly flat (<3%)	Curvature has no effect on vehicle running costs
Rolling	Generally medium gradient (<4%) with many sags and crests.	Significant curves for at least 30% of the length.
Mountainous	Generally steep gradient (<7%) with many sags and crests	Very sharp curves for at least 30% of the length.

Climatic Region

The specific climatic region is indicated according Meteorological Services of Zimbabwe. In specific areas, provision should be made for localized “very wet areas” Climatic Regions are given in the Table below.

Code	Climatic Region
D	Dry
M	Moderate
W	Wet
V	Very Wet

5.4 Geometric Data

Item #	Description	Category		
		Mandatory	Essential	Desirable
1	Chainage from(km)			
2	Chainage to (km)			
3	Segment Length (m)			
4	Segment width (m)	yes		
5	Shoulder Width (m)	yes		

Chainage From and Chainage To (Desirable)

As the name suggests, indicate the chainage where the segment start and where it ends -e.g. if from the 5km peg to 10km peg then the chainage from is 5+00 to 10+00.

The data must be entered in the format 0+00, where the 00 before the + represents km and the 00 after the + represents m. For example at a distance of 3600m (or 3km and 600m) from the datum point, the chainage must be entered as 3+600; and for 13045m (13km and 45m) will be entered as 13+045).

Segment Length (km) – (Desirable)

Please indicate the segment length as measured by the Speedometer Odometer of your car. This is essential so that you can approximate the length of your segment relative to what is highlighted on your tablet. This becomes more critical in cases where the road is not digitized prior to the survey.

Applicable: ALL segments

Segment Width (m) – (Mandatory)

The width of each individual segment must be measured. ALL Segments must be measured.

5.5 Condition or Serviceability Data

Narrow Cracks

These includes any cracks which can be transverse, crocodile, longitudinal or block cracks which are less (<) than 3mm in width.

Applicable: Only on Sealed Roads

Wide Cracks

These includes all cracks which could be transverse, crocodile, longitudinal or block cracks which are equal to or more than (=/>) 3mm in width sealed cracks are regarded as wide cracks.

Applicable: Only on Sealed Roads

Potholes/Patches

Patches are repairs to pavement distresses and are an indication of previous distresses, and maintenance activity, patches include patched potholes.

Applicable: Only on Sealed Roads

Rutting

Rutting results from compaction or shear deformation due to the action of traffic and is therefore, only limited to the wheel paths, These are usually associated with the secondary distresses like crocodile cracks.

Applicable: Only on Sealed Roads

Ravelling (Aggregate Loss) - Desirable

Ravelling (Aggregate loss) is the crumbling and loss of the surfacing agregate, usually as a result of the abrasive action of traffic. Tell tale signs of aggregate loss can sometimes be seen at the side of the road.

Applicable: Only on Sealed Roads

Bleeding/Flushing (Desirable)

Bleeding occurs when excess binders omves upwards relative to the aggregares,therefore reducing surface texture depth. The measurement of this distressi s complicated by the pronounced difference in textures obtained in the different forms of new laid surfacings.

Applicable: Only on Sealed Roads

Edge Breaks

This distress is prevalent on un surfaced shoulders, it can also occur on surfaced shoulders, the most common course of this distress is shoulder edge drop

Applicable: Only on Sealed Roads

Roughness on Sealed Roads

This can be described as the unevenness of the surface and it contributes significantly to **poor riding quality** on most ageing roads. It is one of the most important indicators of structural failure.

Roughness on Gravel Roads

Roughness is assessed visually on gravel roads according to the grades in the paragraphs (annex 2.)

This parameter will only have to be estimated, no physical measurements are necessary .

Riding Quality - (Desirable)

The riding quality of a pavement can be defined as the general extent to which the road uses, through the medium of their vehicles, experience a ride that is smooth and comfortable, or bumpy and therefore unpleasant or perhaps unsafe. This is determined by the unevenness of the road profile, the loss of surface or base layer material and uneven pathcing.

Thickness of gravel wearing Course

This, is essence, is not a distress but a condition of the gravel road it is only to be evaluated on gravel roads.

Thickness of Gravel wearing course is assessed by making test holes, by pick, at the beginning and at the end of the segment and at least at one point on every kilometre.

Cross Section

Cross sections are assessed visually by comparing the shape with those illustrated in **annex 1**

Applicable: Only on Gravel Roads

Corrugations on gravel roads

Washboarding or corrugation of roads comprises a series of ripples, which occur with the passage of wheels rolling over unpaved roads at speeds sufficient to cause bouncing of the wheel on the initially unrippled surface. Most studies of washboarding pertain to granular materials, including sand and gravel.

Applicable: On Gravel or Earth Roads

Drainage

The assessor will have to consider side drains, mitre drains and culverts and if they drain water properly. If they are required, but not provided, then the segment will have the worst degree.

Applicable: Only on ALL Road Types (Sealed, Gravel & Earth)

Passability

This parameter will be evaluated on earth roads. Since most earth roads do not have proper drainage structures, alignment and raised formations, they can become impassable during the wet season. An overall assessment of the whole segment will be recorded on this parameter and those kilometres which are not passable will have to be indicated as such, Generally, earth roads are all passable during dry season and may be impassable during the wet season.

Applicable: Only on ALL Road Types (Sealed, Gravel & Earth)

Information to be collected on the following features:-

6 ATTRIBUTE INFORMATION SUMMARY

6.1 Feature Class: - Sealed Roads

The maximum segment length on sealed roads is 5km.

6.1.1 Basic Data

- Route number
- Road Name
- Road Class (Primary, Secondary, Tertiary Reader, Tertiary Access, Urban Arterial, Urban Collector, Urban Access, Industrial
- Road Type (Wide Mat SS, Wide Mat GS, Narrow Mat, Strip)
- Climatic Region (Dry, Moderate, Wet, Very Wet)
- Terrain Type (Flat, Undulating, Mountainous)
- Datum point reference (description)
- Authority Name (RDC, DoR, DDF)

6.1.2 Geometric Data

- No. of Lanes per carriageway (2,3,4)
- Road Length(km)
- Chainage from (km):-
- Chainage To (km):
- Segment Length (Km)
- Road width(m)
- Shoulder Width (m)
- Drainage type(no drain, V-drain, Trapezoidal, Piped Kerb,)

6.1.3 Condition Serviceability Data

- Servitude vegetation (none, light, medium, dense)
- Narrow cracks degree (no cracks, faint cracks, distinct cracks up to 1mm)
- Wide cracks degree (no cracks, cracks 3-5mm, cracks 5-10mm)
- Pothole/patches degree (good, fair, poor,)

- Rutting degree (No rutting <5mm, Discernible 5-15mm, large 15-25mm)
- Edge breaks Degree (No edge break, up to 50mm, 50-100mm break, > 100mm)
- Edge Drop Degree (No edge break, up to 50mm, 50-100mm break, > 100mm)
- Drainage (Good, fair, poor, Eroded)
- Ravelling Degree (None, Minor, Major)
- Riding quality degree (good, fair, poor)
- Road markings (Yes/No)
- Road studs (Yes/No)
- Passability (all year round; dry season only, wet season only, rupture)
- Grid (Bad, Fair, Good)
- photo

6.1.4 Traffic Data

- Daily traffic - (0-50, 51-100,>101)

6.1.5 Design Data

- Year constructed (to sealed standard(Before 1980, 1980-1990, 1990-2000, 2000- 2010)
- Last surface year

6.2 Feature Class: - Gravel Roads

The maximum segment length on gravel roads is 10km.

6.2.1 Basic Data

- Route number
- Road name
- Road Length(km)
- Datum point description
- Road Class (Primary, Secondary, Tertiary Feeder, Tertiary Access, Urban Arterial, Urban Collector, Urban Local, Industrial)
- Authority Name (xxx RDC, MOT, DDF)
- Climatic region (Dry, Moderate, Wet, Very Wet)

- Terrain Type (Flat, Rolling, mountainous)
- Date & time(current)

6.2.2 Geometric Data

- Chainage from (km):-
- Chainage To (km):
- Segment Length (Km)
- Road width(m)
- Drainage type(no drain, V-drain, Trapezoidal)

6.2.3 Condition Serviceability Data

- Cross Section (Good, Fair, Flat, Inverted)
- Servitude vegetation (none, light, medium,dense)
- Gravel thickness (<50mm, 50-100,> 100)
- Corrugations (None, minor, major)
- Riding Quality- degree (good, fair,poor)
- Riding Quality – extent > 75%)
- Drainage condition (good, fair, poor)
- Drainage extent > 75%)
- Potholes degree (none, minor, major)
- Potholes extent > 75%)
- Passability (all year round; dry season only, wet season only, rupture)
- photo

6.2.4 Traffic Data

Daily traffic (no.) (<10, 10-20; >20)

6.2.5 Design Construction Data

- Year of construction
- Age in years
- Last year of re-gravelling

6.3 Feature Class: - Earth Roads

6.3.1 Basic Data

- Road Name

- Route number
- Datum point description
- Road Class (Primary, Secondary, Tertiary Feeder, Tertiary Access, Urban Arterial, Urban Collector, Urban Local)
- Authority Name
- Climatic region
- Terrain Type

6.3.2 Geometric Data

- Chainage from (km):-
- Chainage To (km):
- Segment Length (Km)
- Road width(m)
- Drainage type(no drain, V-drain, Trapezoidal)

6.3.3 Condition Serviceability Data

- Drainage condition Degree (good, fair, poor)
- Drainage condition extent (1:<25%, 2:25-50%, 3:50-75% 4:> 75%)
- Passability (all year round; dry season only, wet season only, rupture)
- Photo

6.3.4 Traffic Data

- Daily traffic (no.) (< 10, 10-20, >20)

6.3.5 Design Construction Data

- Year constructed

6.4 Feature Class: - Bridges

6.4.1 Basic Data

- Road Name
- Route number
- Date and Time

6.4.2 Geometric Data

- Chainage (km):-
- No. of spans(1, 2,3,4 or more)

- Barricades (headwall, rails,) none
- Erosion Protection (none, stone pitching, gabions, rip-rap)

6.4.3 Condition Serviceability Data

- Structural Condition (intact, minor cracks, major cracks, accident damages)
- Pier/Abutment condition (intact, eroded)
- Approaches (good, fair, poor)
- Serviceability condition(No silt, partly silted, silted)
- Waterway condition (clear, partially blocked, blocked)
- Evidence of overtopping (yes, no)
- Bridge Sign (Yes, No)
- (Headwalls/Rails (None, Good, Fair, Poor)
- photo

6.4.4 Traffic Data

- average daily traffic (no.)

6.4.5 Design Construction Data

- Year constructed

6.5 Feature Class: - Pipe Culverts

6.5.1 Basic Data

- Route number
- Date and Time

6.5.2 Geometric Data

- Chainage (km):-
- Diameter(mm)
- No. of barrels (1, 2,3,4,4 or more)
- Length (m)
- Cover (mm)
- Headwalls (yes, no)
- Culvert markers (0,1,2,3,4)

6.5.3 Condition Serviceability Data

- Structural Condition (intact, minor cracks, major cracks, accident damaged)

- Approaches (good, fair, poor)
- Silt – (Open, partly blocked, Fully blocked)
- Evidence of overtopping (yes, no)
- Headwalls condition (poor, fair, good)
- Photo

6.6 Feature Class: - Box Culverts

6.6.1 Basic Data

- Route number
- Date and Time

6.6.2 Geometric Data

- Chainage (km):-
- Configuration – (single, multiple)
- No. of openings (1,2,3, 4 or more)
- Length along road (m)
- Width across road (m)
- Height (m)
- Cover (mm)
- Headwalls (yes, no)

6.6.3 Condition Serviceability Data

- Structural Condition (good, fair, poor)
- Headwalls (none, poor, fair, good)
- Blockage (Not silted, partially silted, fully blocked)
- photo

6.7 Feature Class: - Shelverts & Armco Pipes

6.7.1 Basic Data

- Route number
- Date and Time

6.7.2 Geometric Data

- Chainage (km):-
- No. of openings (1, 2,3)

- Length along road (m)
- Base Width (m)
- Height (m)

6.7.3 Condition Serviceability Data

- Structural Condition (good, fair, poor)
- Blockage (Not silted, partially silted, fully blocked)
- Photo

6.8 Feature Class: - Piped Causeways

6.8.1 Basic Data

- Route number
- Date & Time (current)

6.8.2 Geometric Data

- Chainage (km):-
- No. of openings (1, 2,3,4 or more)
- Pipe diameter (mm)
- Length along road (m)
- Width across road (m)
- Height (m)
- Cover (mm)
- Headwalls/Guidestones (yes, no)

6.8.3 Condition Serviceability Data

- Structural Condition (intact, minor cracks, major cracks)
- Approaches (good, fair, poor)
- Blockage(Not blocked, partial blockage, full blockage)
- Photos

6.8.4 Design Construction Data

- Year of construction

6.9 Feature Class: - Drift

6.9.1 Basic Data

- Route number

- Authority Name
- Date& Time

6.9.2 Geometric Data

- Chainage (km):-
- Length (m)
- Width (m)
- Guidestones (yes, no)

6.9.3 Condition Serviceability Data

- Structural Condition (intact, minor cracks, major cracks)
- Approaches (good, fair, poor)
- Passability (all weather, dry season only, wet season only)
- Photo

6.10 Feature Class: - Signs

6.10.1 Basic Data

- Route number
- Date and Time

6.10.2 Geometric Data

- Chainage (km):-
- Sign Name(Danger Warning, Informative, Regulatory, Traffic Lights, Other)
- Sign photo

6.10.3 Condition Serviceability Data

- Condition (fair ,poor, aged)
- photo

6.11 Feature Class: - Junctions

6.11.1 Basic Data

- Route number
- Date & Time

6.11.2 Geometric Data

- Chainage (km):-

- Junction Type(T junction, Y junction(merge or diverge), +Intersection, Roundabout, Grade Separated)
- Kerbs (Yes/No)

6.11.3 Condition Serviceability Data

- Sight Lines (Good, fair, poor)
- Photo

6.12 Feature Class: - Grid

6.12.1 Basic Data

- Route number
- Date & Time

6.12.2 Geometric Data

- Chainage (km):-

6.12.3 Condition Serviceability Data

- Structural condition (Good, fair, poor)
- Photo (optional)

6.13 Feature Class: - Impassable Section

6.13.1 Basic Data

- Route number

6.13.2 Geometric Data

- Chainage (km):-

6.13.3 Condition Serviceability Data

- Description
- Intervention Measure
- Photo (optional)

6.14 Traffic Lights

6.14.1 Basic Data

- Route number
- Date & Time

6.14.2 Geometric Data

- Chainage (km):-

6.14.3 Condition Serviceability Data

- Power Source (Solar, ZESA; Hybrid)
- Status (Functional, Non Functional)

6.15 Streetlights

6.15.1 Basic Data

- Route number
- Date & Time

6.15.2 Geometric Data

- Chainage (km):-

6.15.3 Condition Serviceability Data

- Power Source (Solar, ZESA)
- Status (Functional, Non Functional)

6.16 Bus Stop

6.16.1 Basic Data

- Route number
- Date & Time

6.16.2 Geometric Data

- Chainage (km):-

6.16.3 Condition Serviceability Data

- Condition (Good Fair, Poor)

6.17 Layby

6.17.1 Basic Data

- Route number
- Date & Time

6.17.2 Geometric Data

- Chainage (km):-

6.17.3 Condition Serviceability Data

- Condition (Good Fair, Poor)
- Sign Present (Yes, No)

6.18 Rail Crossing

6.18.1 Basic Data

- Route number
- Date & Time

6.18.2 Geometric Data

- Chainage (km):-

6.18.3 Condition Serviceability Data

- Boomgate (Yes, No)
- Safety Barricades (Yes,No)
- Sign presence(yes, No)

6.19 Tollgates

6.19.1 Basic Data

- Route number
- Date & Time

6.19.2 Geometric Data

- No. of lanes (2,4,6)

6.19.3 Condition Serviceability Data

- Condition(good, fair, poor)

6.20 Footbridges

6.20.1 Basic Data

- Route number
- Date & Time

6.20.2 Geometric Data

- Chainage (km):-

6.20.3 Condition Serviceability Data

- Condition(good, fair, poor)

6.21 Settlements

6.21.1 Basic Data

- Route number
- Date & Time

6.22 General Point

This is for capturing any other data that the Assessor may find relevant but cannot be in any of the descriptions provided in this Manual.

6.22.1 Basic Data

- Route number
- Date & Time

6.23 Missing Sealed Road

For Sealed roads which are not digitized (hence missing), or wrongly digitized as Gravel or Earth Roads. Data to be entered is similar to Sealed Roads, but entered as Point Features.

6.24 Missing Gravel Roads

For gravel or earth roads which are not digitized (hence missing on the road map but available physically on the ground), or wrongly digitized as Sealed Roads. Data to be entered is similar to Gravel or Earth Roads. The only difference is that this information is , entered as Point Features.

PART II : ANNEXURES

7 ANNEXURES